

主播话术 (claim)

这外套保暖性堪比羽绒服

商品事实 (evidence)

含绒量: 0%
主面料: 涤纶

(p, a) = (外套#A1, 保暖性)

$\{(X_{ci} X_{ei} | y_i, C_i)\}_{i=1}^N$
training mini-batch

trainable

frozen

positive

hard-neg

arrow color = vector identity

Shared Encoder

BGE-large-zh-v1.5

LoRA (r=16)

x 24

parameters shared across
batches and the two streams

Layer detail: Self-Attn → Cross-Attn → FFN

TwoStreamFusion (x N=2)

A

bidirectional cross-attn (shared QKV)

FAISS index G

partitioned by (attribute, label)

Pseudo-positive

g^+
same attr · same label

Hard negative

g^-
same attr · opposite label

$= -\log \frac{\exp(g \cdot g^+ / \tau)}{\exp(g \cdot g^+ / \tau) + \sum_{g^- \in G} \exp(g \cdot g^- / \tau)}$

attribute-blocked

$L_{total} = L_{CE} + \lambda \cdot L_{CL}$

Logistic Classifier

$-\log \hat{y} \cdot (1 - y) - (1 - \hat{y}) \log (1 - \hat{y})$

reliability-weighted binary cross-entropy

MLP → L2-Norm

$g_{p,a} \in \mathbb{R}^{56}$

RKC (k-NN)

retrieval-augmented

$|\hat{y} - \hat{y}_{RKC}| < \delta ?$

① Inputs ② Shared encoder + LoRA

③ TwoStreamFusion

④ Heads · contrastive learning · inference